Forces

Ch 5 HRW

Newton's 1^{st}, 2^{nd} and 3^{rd} law

Specific forces

Force diagrams

Newtonian Mechanics

- Speeds of the interacting bodies is much less than the speed of light.
- Objects are in the size range from very small (almost sub-atomic) to very large (galaxies).
- Before Newton:
  - To keep an object moving, it was necessary to apply a force to maintain constant velocity.
  - Natural state = at rest.
- Friction was not taken into account

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Newton's 1st Law

- Newton's First Law: If a body is at rest, or is moving with constant velocity, it will continue to do so if no **NET** force acts on it.
- The object continues to move with constant velocity (magnitude and direction)
- $$\vec{v} = \vec{v}_0$$
  $$\Delta\vec{v} = \vec{v} - \vec{v}_0 = 0$$
  $$\vec{a} = \frac{d\vec{v}}{dt} = 0$$
- There is no acceleration if: $\vec{F}_{net} = ma = \vec{F}_1 + \vec{F}_2 + ... = 0$
  $$\therefore \vec{F}_{net} = \sum \vec{F} = ma = 0$$

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Force

- A force is measured by the acceleration it produces.
- Force is a vector quantity (Magnitude and direction)
- Read section 5-4
- More than one force $\Rightarrow$ resultant (net) force
- Inertial frames – one in which Newton's laws hold.

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Mass

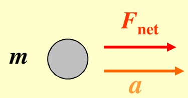
- A given force will produce different accelerations for different objects.
- Kick/throw a soccer ball and a bowling ball with the same force – what are the effects?
- The mass of a body is the characteristic that relates the force on the body to the resulting acceleration.

Newton's 2nd Law

- Newton's 2nd law: The <u>NET</u> force on a body is equal to the product of the body's mass and its acceleration.

$$\vec{F}_{net} = m\vec{a}$$

- The direction of the net force is in the same direction as the acceleration



- Need to consider the NET force – i.e. the vector sum of all the forces acting on the body.
- Forces must act ON THAT BODY, not another body.

Newton's 2nd Law

- Forces must act ON THAT BODY, not another body.
- Always consider

$$F_{net,x} = ma_x \quad , \quad F_{net,y} = ma_y \quad , \quad F_{net,z} = ma_z$$

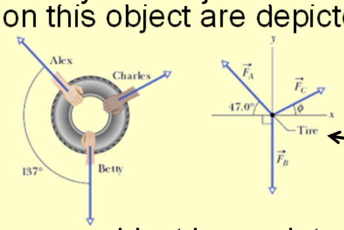
- If net force = 0 N, then ALL components must also = 0 N i.e.

If $\vec{F}_{net} = \sum \vec{F} = 0$

Then $F_{net,x} = ma_x = 0 \quad , \quad F_{net,y} = ma_y = 0 \quad , \quad F_{net,z} = ma_z = 0$

Newton's 2nd Law

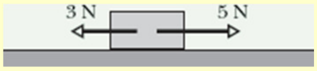
- Equilibrium –there is no net force.
- Force diagram: only one object is shown, and all the forces acting on this object are depicted on the force diagram.



- Free body diagram: object is a point and the forces are drawn from the origin of the co-ordinate system.
- External force: in a system of 1 or more bodies, any force from outside the system on the bodies is an external force.
- Internal force: we do not include these in the FB diag.

Checkpoint 2

- The figure here shows two horizontal forces acting on a block on a frictionless floor. If a third horizontal force $\vec{F}_3$ also acts on the block, what are the magnitude and direction of $\vec{F}_3$ when the block is (a) stationary and (b) moving to the left with a constant speed of 5 m/s?



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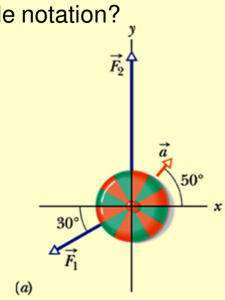
Sample Problem p94

- In the overhead view of a 2.0 kg box is accelerated at 3.0 m/s^2 in the direction shown by $\vec{a}$, over a frictionless horizontal surface. The acceleration is caused by three horizontal forces, only two of which are shown: $\vec{F}_1$ of magnitude 10 N and $\vec{F}_2$ of magnitude 20 N. What is the third force $\vec{F}_3$ in unit-vector notation and in magnitude-angle notation?



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Sample Problem p94

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Some Particular Forces

- Gravitational Force: It is the force that the Earth exerts on any object (in the picture a cantaloupe). It is directed toward the center of the Earth. Its magnitude is given by Newton's second law.

$$F_{net,y} = ma_y$$
$$-F_g = m(-g)$$
$$F_g = mg$$
$$\vec{F}_g = -F_g\hat{j} = -mg\hat{j}$$

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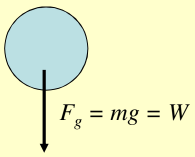
Some Particular Forces

- Weight: $\neq$ mass
- Weight (W): The weight of a body is equal to the magnitude F_g of the gravitational force on the body.
  - If there are two objects, suppose the gravitational force on the one is 2.0 N and the other one is 3.0 N. Then the second one is said to be heavier than the first.
-



- If the object is moved to a location where the acceleration of gravity is different (e.g., the moon, where g_m = 1.7 m/s^2), the mass does not change but the weight does.

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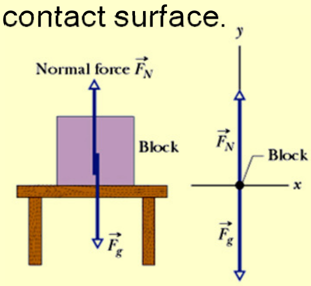
Some Particular Forces

- Normal Force: When a body presses against a surface, the surface deforms and pushes on the body with a normal force perpendicular to the contact surface.
- Symbol is: $\vec{F}_N$ or $\vec{N}$
- It is always perpendicular to the surface on which the object lies.
- Example – only 2 forces experienced by the block – no acceleration
- In this case: $F_{net,y} = F_N - F_g = ma_y$

$$F_N = mg + ma_y = m(g + a_y)$$

$$F_N = mg$$



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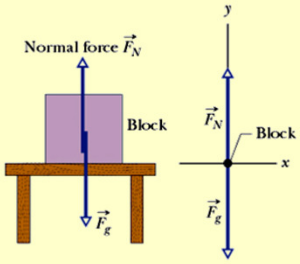
Checkpoint 3

- Is the magnitude of the normal force greater, less than or equal to mg if the block and table are in a lift moving upward
  (a) at constant speed?
  (b) at increasing speed?



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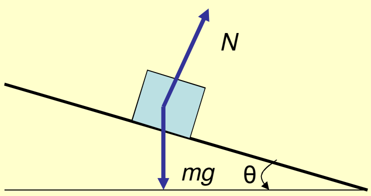
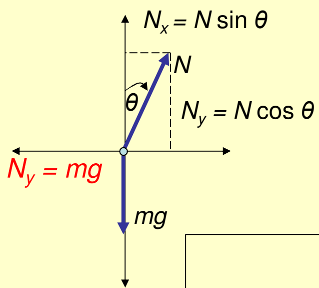
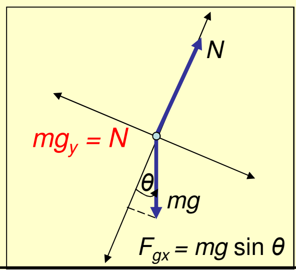
Some Particular forces

- Normal force – stationary object on a slope





$N \neq mg$



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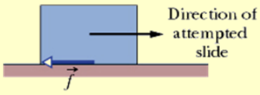
Some Particular Forces

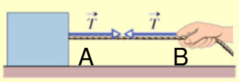
- Friction: If we slide or attempt to slide an object over a surface, the motion is resisted by a bonding between the object and the surface. This force is known as "friction."
- Symbol usually: $\vec{f}$
- Friction is always in the opposite direction to motion



- Many problems state that motion is on a frictionless surface.

Some Particular Forces

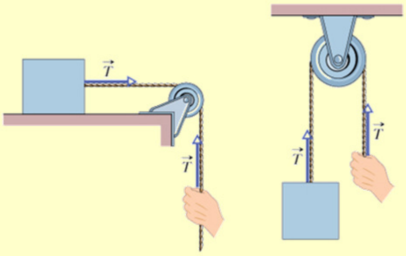
- Tension: This is the force exerted by a rope or a cable attached to an object
- Tension has the following characteristics:
  - It is always directed along the rope.
  - It is always pulling the object.
  - It has the same value along the rope (for example, points *A* and *B*).



- The following assumptions are made:
  - The rope has NEGLIBIBLE MASS compared to the mass of the object it pulls.
  - The rope does NOT STRETCH.

Some Particular Forces

- Tension: This is the force exerted by a rope or a cable attached to an object
- If a pulley is used as in the figure, we assume that the pulley is massless and frictionless (in the following 2 chapters.
- Remember the tension is the same along the length of the rope.



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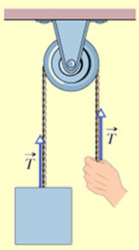
Checkpoint 4

- The suspended body in the figure weighs 75 N. Is T equal to, greater than or less than 75 N when the body is moving upward
  (a) at constant speed,
  (b) at increasing speed, and
  (c) at decreasing speed?



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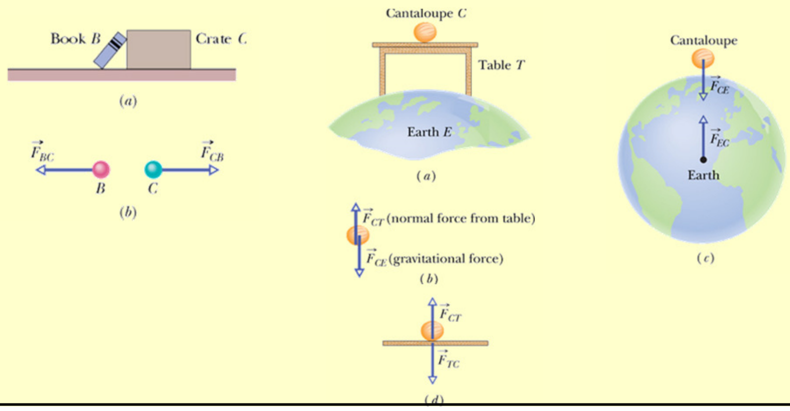
Newton's 3rd Law

- Newton's 3rd Law: When two bodies interact by exerting forces on each other, the forces are equal in magnitude and opposite in direction.



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NB NOTE

- The rest of this chapter consists of sample problems. You should make sure you understand ALL of them, learning their procedures for attacking a problem. Especially important is knowing how to translate a sketch of a situation into a free-body diagram with appropriate axes, so that Newton's 2nd law can be applied.

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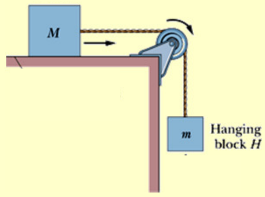
Sample Problem p100

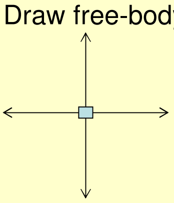
Read again carefully in HRW

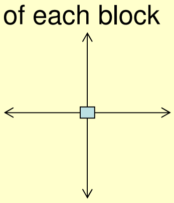
- The Figure shows a block S (the sliding block) with mass M = 3.3 kg. The block is free to move along a horizontal frictionless surface and connected, by a cord that wraps over a frictionless pulley, to a second block H (the hanging block), with mass m = 2.1 kg. The cord and pulley have negligible masses compared to the blocks (they are “massless”). The hanging block H falls as the sliding block S accelerates to the right. Find (a) the acceleration of block S, (b) the acceleration of block H, and (c) the tension in the cord.
- What is the problem about?
  - 2 bodies (blocks) with influence of Earth (gravity)
  - Identify all the forces on the blocks.
  - Stretchless cord
  - Blocks move together
  - Blocks have same acceleration a
  - Draw free-body diagram of each block



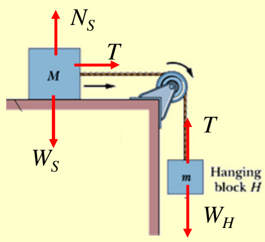
Sample Problem p100

- M = 3.3 kg m = 2.1 kg
  Find (a) the acceleration of block S, (b) the acceleration of block H, and (c) the tension in the cord.
- Draw free-body diagram of each block



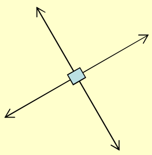


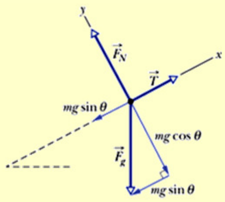
- Apply Newton2 to both blocks separately.

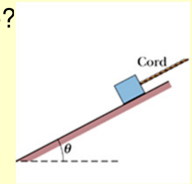


Sample Problem p102

- In Fig. 5-16a, a cord pulls on a box up along a frictionless plane inclined at θ = 30°. The box has mass m = 5.00 kg, and the force from the cord has magnitude T = 25.0 N. What is the box's acceleration component a along the inclined plane?







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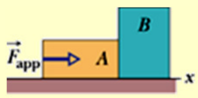
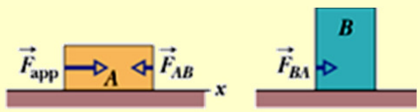
Sample Problem p104

- In the Figure, a constant horizontal force of magnitude 20 N is applied to block A of mass m_A = 4.0 kg, which pushes against block B of mass m_B = 6.0 kg. The blocks slide over a frictionless surface, along an x axis. (a) What is the acceleration of the blocks? (b) What is the (horizontal) force on block B from block A?



(a)



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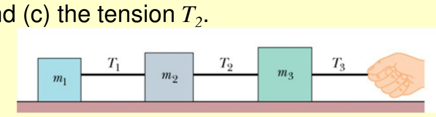
Problem 53

- In the figure, three connected blocks are pulled to the right on a horizontal frictionless table by a force of magnitude T_3 = 65.0 N. If m_1 = 12 kg, m_2 = 24.0 kg and m_3 = 31.0 kg, calculate (a) the magnitude of the system's acceleration, (b) the tension T_1 and (c) the tension T_2.



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